Long Term Strategy

Which equity income strategy long-term investors should adopt?

- The current regime of "2020–2026 Post-COVID — AI era" has posed a challenge for traditional dividend income funds targeting high dividend yield as they significantly underperformed the broader market.
- As a result of this large underperformance, there has been an increased focus on three alternative approaches beyond traditional high dividend yield strategies:
  - Targeting high dividend growth stocks
  - Targeting high total shareholder yield stocks (i.e. dividend plus buyback yield)
  - Enhancement of equity income strategies via call overwriting i.e., strategies that harvest options income by sacrificing some of the equity upside
- With the AI hype likely to eventually normalise, the regime we are likely to be on over the next ten years is in our opinion more likely to be like the "1963–1989 Normal business cycle swings" rather than the current "2020–2026 Post-COVID — AI era" regime.
- This implies that over the long-term, the next ten years or so, asset allocation should factor in a strong positive correlation between the dividend and the value factor, a strong negative correlation between the dividend and the market/beta factor and a strong negative correlation between the dividend and the profitability factor.
- At same time given the attractiveness of dividend growth strategies in terms of offering the best balance of good vol-adjusted returns, relatively consistent overall performance in drawdowns and some exposure to tech/growth factor, we believe that long-term investors aiming for equity income going forward should combine traditional dividend yield strategies with dividend growth strategies.

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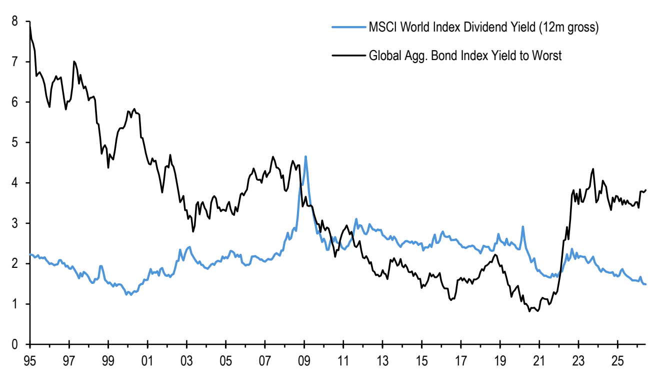
Equity income strategies such as those focusing on dividend yields have traditionally been a significant component of investor portfolios as historically around a third of total equity returns have come from dividends. The importance of equity income strategies naturally increases in periods of low interest rates such as that seen between the financial crisis of 2008 and 2021.

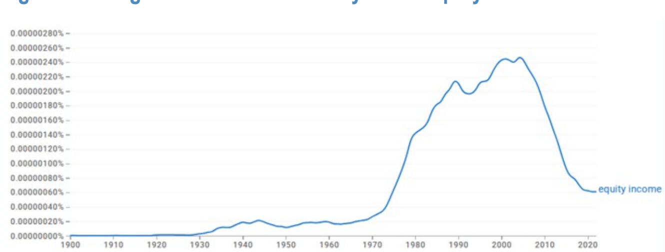
Since 2022, the shift to a higher interest rate environment has somewhat hit the relative attractiveness of equity dividend yields vs bond yields. However, as shown by Figure 1 which depicts the MSCI World Index dividend yield vs the Global Agg Bond Index yield to worst, the gap between bond yields and dividend yield had been much larger in previous periods, such as during 1990s, without denting the interest on high dividend yield stocks at the time. This is shown by Figure 2 which depicts the google books search for keyword “equity income”. The frequency of mentioning or popularity of the theme of “equity income” kept rising steadily from 1970s to around 2004 with a declining trend from 2004 till the pandemic. Post the pandemic there has been stabilization.

Figure 1: MSCI World Index dividend yield vs Global Agg Bond Index yield to worst



Source: Bloomberg Finance L.P., J.P. Morgan.

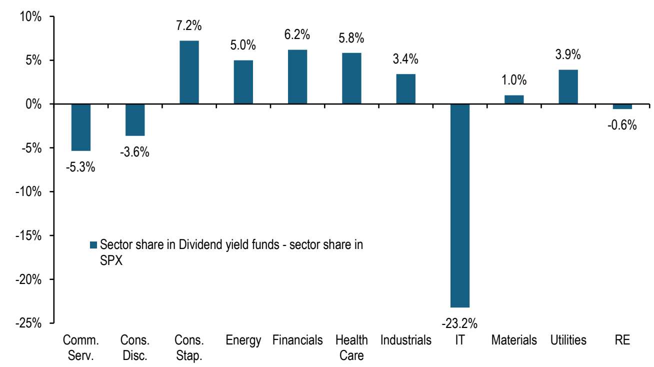
Figure 2: Google books search for keyword “equity income”



Source: Google.

The decline since 2004 perhaps reflects the rise in tech in equity markets which reduced the attractiveness of traditional dividend income strategies given these strategies are inherently underweights tech. This is shown in Figure 3, which depicts the sectoral deviation vs the S&P500 index of the 10 largest US high dividend yield funds. These high dividend yield funds have been typically UW Tech, Communication Services and Consumer Discretionary sectors and OW Financials, Staples, Energy, Healthcare.

Figure 3: Sectoral deviation vs the S&P500 index of the 10 largest US high dividend yield funds



Source: Bloomberg Finance L.P., J.P. Morgan.

That said, high dividend yield stocks have other properties that make them attractive to investors. For example using the Fama-French equity factors and by looking at the correlation matrix (Figure 4) of 6 factors (Mkt_RF-overall market or beta factor, SMB-size factor, HML-value factor, RMW-operating profitability factor, CMA-conservative investment factor, DY-dividend yield factor proxied by the difference in the performance of high vs low dividend yield stocks), we find that there has been strong negative correlation between the market/beta factor and the dividend yield factor, suggests that high dividend yield stocks historically served as a hedge to equity market drawdowns. The strong negative correlation between the market factor and the dividend factor reflects at least partly the growth tilt of the equity market as during strong equity market phases, growth stocks tend to outperform more value oriented /dividend stocks.

Figure 4: Correlation Matrix

Based on monthly data from 1963 to 2026.

	Mkt_RF	SMB	HML	RMW	CMA	DY
Mkt_RF	1	0.27	-0.21	-0.19	-0.36	-0.52
SMB	0.27	1	0.01	-0.34	-0.08	-0.21
HML	-0.21	0.01	1	0.09	0.68	0.68
RMW	-0.19	-0.34	0.09	1	0.01	0.06
CMA	-0.36	-0.08	0.68	0.01	1	0.61
DY	-0.52	-0.21	0.68	0.06	0.61	1

Source: J.P. Morgan.

The correlation matrix of Figure 4 provides some additional insights:

- Value–Conservative Investment–Dividend Cluster (HML, CMA, DY factors) are highly correlated and thus using all three of them in a multi-factor portfolio would induce multicollinearity. What creates this high correlation is relatively simple: firms with high book-to-market (value stocks) tend to be conservative in terms of their propensity to invest and firms that invest conservatively tend to also pay higher dividends.

- The profitability factor (RMW) is effectively orthogonal to the value/conservative investment/dividend cluster.
- The size factor is negatively correlated with the dividend factor as small stocks tend to pay lower dividends.

Do the above correlations change across sub periods? Have there been any structural changes? To answer this question, we look at Figure 5 which depicts the DY factor correlations across sub-periods.

Figure 5: DY Factor Correlations Across Sub-Periods

Period	Mkt_RF	SMB	HML	RMW	CMA
1963-1975	-0.50	-0.23	0.80	-0.49	0.75
1976-1989	-0.69	-0.41	0.80	-0.31	0.62
1990-1999	-0.66	-0.32	0.74	0.21	0.75
2000-2009	-0.34	-0.26	0.68	0.42	0.50
2010-2019	-0.60	-0.31	0.07	0.47	0.37
2020-2026	-0.30	0.39	0.76	0.08	0.58

Source: J.P. Morgan.

Overall Figure 5 points to five distinct regimes:

1963–1989 (Normal business cycle swings): DY strongly correlated with HML (+0.80) and CMA (+0.75), strongly negatively correlated with Mkt_RF (−0.50 to −0.68) and RMW (−0.31 to −0.49). Traditional correlation between “value” and “income” factors.

1990–1999 (Dot-Com bubble era-dominance of growth stocks): DY vs RMW flips from negative to positive (+0.21). As growth stocks dominated, the profitability factor’s relationship with dividends changed.

2000–2009 (credit bubble and GFC crisis): DY vs RMW reaches +0.42. Flight-to-quality during the financial crisis strengthened the link between dividends and profitability. DY vs Mkt_RF weakens to −0.34.

2010–2019 (Low-Rate Environment post GFC): structural break, DY vs HML collapses to +0.07 (from +0.68 in prior period). The prolonged low-rate environment caused investors to chase yield regardless of value characteristics, decoupling dividend yield from traditional value.

2020–2026 (Post-COVID — AI era): Massive policy interventions and the emergence of the AI trade reshape factor relationships, DY vs SMB flips positive (+0.39) for the first time in 60 years. DY vs HML rebounds to +0.76. DY vs RMW drops back to near-zero (+0.08).

The above observations suggest portfolio construction/asset allocation for long term investors is regime dependent. And with the AI hype likely to eventually normalise, the regime we are likely to be on over the next ten years is in our opinion more likely to be like the “1963–1989 Normal business cycle swings” rather than the current “2020–2026 Post-COVID — AI era” regime.

This implies that over the long-term, the next ten years or so, asset allocation should factor in a strong positive correlation between the dividend and the value factor, a strong negative correlation between the dividend and the market/beta factor and a strong negative correlation between the dividend and the profitability factor.

At the same time, we recognize that the current regime of “2020–2026 Post-COVID — AI era” has posed a challenge for traditional dividend income funds targeting high dividend yield as they significantly underperformed the broader market. As a result of this large underperformance, there has been an increased focus on three alternative approaches beyond traditional high dividend yield strategies:

1. Targeting high dividend growth stocks
2. Targeting high total shareholder yield stocks (i.e. dividend plus buyback yield)
3. Enhancement of equity income strategies via call overwriting i.e., strategies that harvest options income by sacrificing some of the equity upside

How big has the shift been towards these three new approaches? Have they outperformed the traditional approach targeting high dividend yields?

To assess how these strategies have performed in practice, we look at total returns on equity ETFs and mutual funds in the following categories: dividend yield-focused, dividend growth-focused, total shareholder yield and enhanced or option-augmented equity income. Returns are equal-weighted within each category to reflect strategy-level outcomes, rather than be dominated by individual large funds, and we focus on the period from 2014 onward, as there are few funds of the latter three categories trading before then. We use up to ten of the largest ETFs and mutual funds for each category. Figure 6 shows the cumulative returns for the four strategies. While the four strategies had followed each other more closely up to 2020, there has been a notable divergence since then with the

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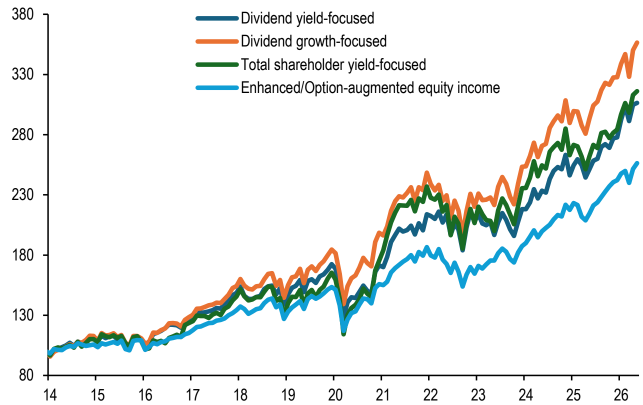
dividend growth strategies compounding most strongly.

Figure 6: Cumulative return on equity income funds



Source: J.P. Morgan.

Figure 7 shows annualized returns and information ratios for these strategies for the full sample as well as a split to pre-2020 and post-2020 periods. The overall returns and information ratios for the traditional income strategies focused on dividend yield and dividend growth were closer to the S&P 500 in the pre-2020 period. Post-pandemic, however, the returns of the S&P, increasingly dominated by a relatively small number of mega-cap tech companies that are underweighted by these income strategies, have outperformed both outright and on a risk-adjusted basis. Dividend growth funds outperformed dividend yield funds in both periods both on an outright and risk-adjusted basis, but while the total yield focused funds outperformed both in the post-2020 period outright higher volatility dragged risk-adjusted returns lower. The latter may be e.g. due to higher tech content and a smaller, less established fund universe of total yield focused funds. And the enhanced/option-augmented strategies have produced similar returns on a risk adjusted basis post-2020 at the expense of somewhat lower absolute returns, likely on a combination of a reduction in market beta via premium income from call overwriting offsetting some of the negative returns in drawdowns (and some strategies within these category buy puts for downside protection) while also capping some of the upside in recoveries.

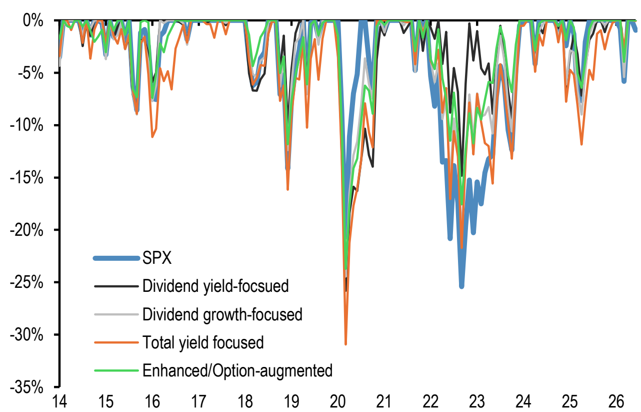
Figure 7: Annualized return and information ratio of S&P 500 and equity income funds

Annualized return	SPX	Dividend yield-focused	Dividend growth-focused	Total yield focused	Enhanced/ Option-augmented
2014-2019	11.3%	9.6%	10.9%	9.2%	7.6%
2020-2026	14.8%	10.3%	11.7%	12.1%	8.9%
Full sample	13.0%	10.0%	11.4%	10.6%	8.3%
IR					
2014-2019	1.00	0.94	0.97	0.71	0.83
2020-2026	0.86	0.64	0.70	0.62	0.68
Full sample	0.89	0.74	0.80	0.64	0.74

Source: J.P. Morgan.

To examine drawdowns more specifically, Figure 8 shows cumulative maximum drawdowns from prior peaks in time series form, while Figure 9 shows the maximum end-month drawdown for each strategy across six key correction episodes.

Figure 8: Drawdowns of S&P 500 and equity income fund types



Source: J.P. Morgan.

Figure 9: Maximum drawdowns by episode

Maximum end-month drawdown in:	SPX	Dividend yield-focused	Dividend growth-focused	Total yield-focused	Enhanced/ Option-augmented
Sep15/Jan16 China deval/EM growth scare	-8.6%	-8.7%	-9.0%	-11.1%*	-7.3%
Dec18 Fed tightening/liqudiity withdrawal	-14.2%	-10.1%	-12.4%	-16.2%	-11.8%
Mar20 Pandemic	-20.7%	-25.8%	-24.5%	-30.9%	-23.7%
Sep22 Inflation shock	-25.4%	-14.8%	-20.6%	-21.7%	-17.6%
Apr25 Liberation day	-7.7%	-7.2%	-9.0%	-11.9%	-6.4%
Mar26 Middle East conflict	-5.8%	-4.1%	-5.4%	-2.6%	-4.0%

* Trough in Jan16 for total yield-focused, Sep15 for S&P and other income strategies.
Source: J.P. Morgan.

The first episode shows a relatively uniform drawdown across the S&P as well as different equity income types. The second shows significant downside mitigation relative to the broader market for most income strategies (with the exception of total shareholder yield), particularly for the dividend yield focused and option-augmented income strategies, where a value/short duration tilt supported the former while the latter saw offsets from call writing to augment income and/or buy some puts for downside protection. The third, the pandemic shock, hit equity income strategies harder than the broader market both on the way down and in the recovery as earnings impairments and dividend cuts in energy, financials and real estate weighed on income strategies disproportionately while the S&P's greater exposure to tech supported the broader market on a relative basis. The fourth was an inflation shock that prompted significant central bank interest rate hikes, which disproportionately hurt long-duration growth stocks and within income strategies dividend growth focused funds with a higher quality/growth tilt that also skews longer duration as well as total shareholder yield focused funds amid longer-duration tech/growth tilts than dividend yield strategies. By contrast, income funds focusing on current yields with a shorter-duration tilt held up better and option-augmented strategies perhaps also benefited from higher implied volatility boosting premium income and/or put exposure. The fifth was a tariff shock and the sixth a geopolitical uncertainty shock that both saw elevated uncertainty, which saw broadly similar drawdowns overall though the option-augmented income funds likely benefited from option premium income while total shareholder yield did better in the sixth episode.

Taken together, the above highlights that the relative drawdown performance of equity income funds is fundamentally regime-dependent. In monetary tightening-driven corrections in 2018 and 2022, income strategies particularly based on dividend yield benefited from a short-duration value tilt while long duration assets suffered. In the earnings/cash flow destruction shock in 2020, income strategies broadly underperformed the broader market. Dividend yield and growth strategies were penalised as their sector composition (e.g. lower exposure to tech and higher exposure to sectors where earnings/dividends suffered) amplified the initial loss and lagged in the recovery, while total shareholder yield strategies likely suffered from higher mid-cap exposure. And in the shorter, sharper vol spikes in 2015, 2025 and 2026 option-augmented strategies most consistently demonstrated resilience from premium income. A related distinction is that traditional income strategies based on dividend yield and dividend growth can behave differently, with the former having more of a value tilt given it focuses on current yield while the latter has more of a quality/growth tilt, while total shareholder yield strategies appear to introduce a more procyclical element via buybacks.

In the shorter, sharper volatility episodes of 2015, 2025, and 2026, option-augmented strategies most consistently demonstrated resilience from premium income, while total shareholder yield strategies were disproportionately impaired in the tariff shock of 2025 — reflecting their greater multinational technology and industrial exposure — but proved relatively resilient in the geopolitical shock of 2026. A related distinction is that the traditional income strategies themselves differ meaningfully in their factor tilts: dividend yield strategies carry a more pronounced value orientation reflecting their focus on current income; dividend growth strategies have a higher quality and growth tilt; and total shareholder yield strategies sit closer to the broader market in factor space but introduce a procyclical element through their buyback exposure that can amplify drawdowns in liquidity and earnings stress episodes.

Finally, we look at the correlations of the excess returns of the four equity income strategy variants over S&P 500 returns to the Fama-French factors as well as to interest rates and how they differ between dividend yield focused, dividend growth focused, total shareholder yield and enhanced/option-augmented income strategies. This is shown in Figure 10.

Figure 10: Correlations of equity income strategy excess returns over S&P 500 with Fama-French factors

Correlation coefficient	Mkt-RF	SMB	HML	RMW	CMA	DY
Dividend yield-focused						
2014-2019	-0.48	-0.05	0.50	0.32	0.73	0.81
2020-2026	-0.38	0.29	0.78	0.21	0.67	0.88
Full sample	-0.39	0.21	0.72	0.23	0.67	0.86
Divdiend growth-focused						
2014-2019	-0.15	0.11	0.41	0.45	0.47	0.46
2020-2026	-0.23	0.45	0.75	0.26	0.59	0.80
Full sample	-0.21	0.36	0.68	0.28	0.56	0.75
Enhanced/option-augmented						
2014-2019	-0.66	-0.23	-0.12	0.00	0.07	0.29
2020-2026	-0.74	0.03	0.30	0.08	0.33	0.52
Full sample	-0.72	-0.06	0.18	0.06	0.26	0.45
Total yield-focused						
2014-2019	0.27	0.57	0.54	-0.12	0.29	0.00
2020-2026	0.11	0.70	0.78	0.06	0.49	0.72
Full sample	0.14	0.65	0.73	0.02	0.45	0.59

Source: J.P. Morgan.

For dividend yield-focused strategies, the dominant exposures are HML (value) and CMA (conservative investment), consistent with the aim of identifying relatively cheap, mature and conservatively managed companies with high payout ratios that pay high dividend yields. From 2020 onward, the value factor has if anything been stronger, while there has also been some increase in exposure to the SMB factor suggesting some compositional shift in the dividend yield universe toward small caps. The negative market factor loading also suggests it provides genuinely defensive characteristics relative to the broad market, though it appears to have weakened somewhat post-2020. Unsurprisingly, exposure to the dividend/income

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factor is consistently high given the focus of the strategy on dividend yield.

For dividend growth-focused strategies, the key differentiator has been a higher loading to the RMW (quality/profitability), at the expense of a lower DY (income/dividend) factor loading before 2020. That said, post-2020 there has been a greater convergence in many factor loadings with the dividend yield focused funds, which could have some implications for diversification relative to the dividend yield strategies, though it remains less defensive.

For total shareholder yield-focused strategies, the loadings differ from dividend yield or dividend growth focused strategies in notable ways. The correlation of excess returns over the S&P with the market factor is positive, suggesting reliance on buybacks (that are more discretionary in nature) has made the strategy more procyclical than other income strategies. The SMB factor correlation is higher than the other strategies, suggesting a higher small/mid-cap exposure. The RMW loading is the lowest, suggesting little quality exposure, while the value factor loading appears to be more similar to dividend yield strategies. While the DY factor has been low before 2020, this could be affected by the strategy being smaller and perhaps less established.

For enhanced/option-augmented strategies, loadings on most factors have been rather muted, while the market factor is the most negative. Post-2020, there has been more of a loading to value and conservative investment factors, as well as the dividend factor, suggesting that the strategy's income characteristics are more visible. In terms of drawdown behaviour, these strategies have produced the most consistent relative outperformance versus the S&P 500 during broad market corrections, though as the 2020 episode illustrates this does not preclude significant absolute losses.

The above return, drawdown and factor correlation characteristics reinforce the conclusion that there is no single strategy that dominates across different regimes. Instead, they highlight that the decision of allocating to these different income strategies depends on the ultimate aim, and they have come at an opportunity cost in terms of returns relative to the broader market particularly post-2020. For producing current income and a more value/conservative tilt, the dividend yield focused strategies have performed better while providing some downside protection in most of the drawdown episodes in our sample. For a more quality/growth oriented income strategy the dividend growth strategies perform better, though they provide less defensive exposure. Enhanced/option-augmented strategies provide more emphasis on hedging properties vs. the broader market with some exposure that is value/conservative tilted in the post-2020 period, albeit at the expense of lower absolute returns even if risk-adjusted returns are comparable to other income strategies. Total shareholder yield strategies, with a reliance on buybacks that are more discretionary in nature, are more procyclical in nature, providing higher outright returns than other income strategies in the post-2020 period but at the cost of higher volatility that has provided a greater headwind to risk-adjusted returns - a trade-off that the factor analysis confirms is structural.

Figure 11 summarizes some of the key characteristics of the different income strategies taking into account risk-adjusted returns, performance in drawdown episodes as well as tech/growth factor exposure given that income strategies' low tech/growth exposure creates a persistent return drag if tech dominance continues. Of the four groups, the dividend growth strategies appear to offer a balance of good vol-adjusted returns, relatively consistent overall performance in drawdowns and some exposure to tech/growth factor. In principle, dividend yield and enhanced/option-augmented strategies offer better performance in drawdowns, but the former has little tech exposure and while some of the latter have higher tech exposure the effective beta is dampened by call overwriting. And while total yield strategies offer more tech/growth exposure, procyclical character, reflected in the weakest risk-adjusted returns of any income category and poor drawdown performance in most stress episodes, weighs against them as a core income allocation.

Figure 11: Summary table

	Dividend yield-focused	Dividend growth-focused	Total yield focused	Enhanced/option-augmented
Vol-adjusted return	Medium	Good	Poor	Medium
Drawdowns during equity corrections	Good in rate shocks; poor in earnings shocks	Moderate	Poor; procyclical	Good; most consistent overall
Tech/growth factor exposure	Low	Low/moderate	Moderate	Mediun/high notional; Low effective

Source: J.P. Morgan.

In all, with the AI hype likely to eventually normalise, the regime we are likely to be on over the next ten years is in our opinion more likely to be like the "1963–1989 Normal business cycle swings" rather than the current "2020–2026 Post-COVID — AI era" regime. This implies that over the long-term, the next ten years or so, asset allocation should factor in a strong positive correlation between the dividend and the value factor, a strong negative correlation between the dividend and the market/beta factor and a strong negative correlation between the dividend and the profitability factor. At same time given the attractiveness of dividend growth strategies in terms of offering the best balance of good vol-adjusted returns, relatively consistent overall performance in draw-

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